

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

**ENVIRONMENTAL SCIENCE****Paper 2 (Generated practice paper - Set 23)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

**Time allowed: 3 hours****Materials**

For this paper you may use:

- a calculator.

**Instructions**

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

**Information**

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

**For Examiner's Use**

| Question | Mark |
|----------|------|
| 1        |      |
| 2        |      |
| 3        |      |
| 4        |      |
| 5        |      |
| 6        |      |
| 7        |      |
| 8        |      |
| 9        |      |
| 10       |      |
| 11       |      |
| TOTAL    |      |

Answer all questions in the spaces provided.

**0 1**    The Rio Bravo Conservation and Management Area in Belize exploits mahogany under a sustainable management plan while conserving a rainforest of over 200 tree, 400 bird and 70 mammal species.

**0 1** . **1**    Describe **five** features of the timber harvesting at Rio Bravo that make it sustainable.

[5 marks]

- 1 .....
- 2 .....
- 3 .....
- 4 .....
- 5 .....

**0 1** . **2**    Explain why involving local communities is essential to the success of the reserve.

[3 marks]

- .....
- .....
- .....
- .....
- .....
- .....

**0 1** . **3**    Explain why selective logging is better for wildlife than clear-felling, and give **one** disadvantage.

[2 marks]

- .....
- .....

10

0 2 . 1

Explain how the relationship between coral polyps and zooxanthellae allows reefs to support a high biomass in nutrient-poor tropical seas.

[4 marks]

0 2 . 2

Describe the abiotic conditions needed by reef-building corals and explain why these make reefs vulnerable.

[3 marks]

0 2 . 3

Explain **three** ways in which coral reefs are important to humans.

[3 marks]

1 .....

2 .....

3 .....

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**0 3 . 1** Explain how each of the following changes in abiotic factors caused by human activities can make a habitat unsuitable for species living there.

**[6 marks]**

**Water availability** .....

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**Dissolved oxygen** .....

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**pH** .....

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**0 3 . 2** Explain how increased water turbidity caused by ploughing, mining or dredging harms aquatic organisms.

**[2 marks]**

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03 . 3

Explain why species that were already rare are most likely to be lost when abiotic conditions change.

[2 marks]

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04 . 1

Which of the following would be most suitable for monitoring the movements of a whale over several months? Tick one box.

[1 mark]

- A Radio collar ☐
- B Acoustic / satellite tag ☐
- C Pitfall trap ☐
- D Geolocator tag recovered by recapture ☐

04 . 2

Explain why post-release monitoring is important in a captive breeding and release programme and describe **two** suitable methods.

[4 marks]

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**0 5**

Table 1 shows the mean dry mass of wheat plants grown with two fertiliser treatments (10 replicates each).

**Table 1**

| Treatment                | Mean dry mass / g | Standard deviation / g |
|--------------------------|-------------------|------------------------|
| Organic manure           | 14.2              | 3.9                    |
| Inorganic NPK fertiliser | 16.1              | 4.1                    |

**0 5****1**

Explain what the standard deviation values tell you about the data.

**[2 marks]**

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**0 5****2**

Name the statistical test that should be used to decide whether the difference between the two means is significant, and justify your choice.

**[2 marks]**

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**0 5****3**

The test gave a probability  $p = 0.30$ . State what this means and what the student should conclude.

**[2 marks]**

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**0 5****4**

Explain why a statistically significant correlation does not prove a causal relationship.

**[2 marks]**

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0 5 . 5

Suggest how the student could increase the chance of detecting a real difference if one exists.

[2 marks]

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0 6 . 1

Explain why food surpluses produced in MEDCs in the 1970s could not easily be sold.

[3 marks]

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0 6 . 2

Describe **four** methods used to reduce over-production while maintaining farm incomes.

[4 marks]

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06 . 3

Describe **three** features that farmers may be paid for under an agri-environmental scheme and the conservation purpose of each.

[3 marks]

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2 .....

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3 .....

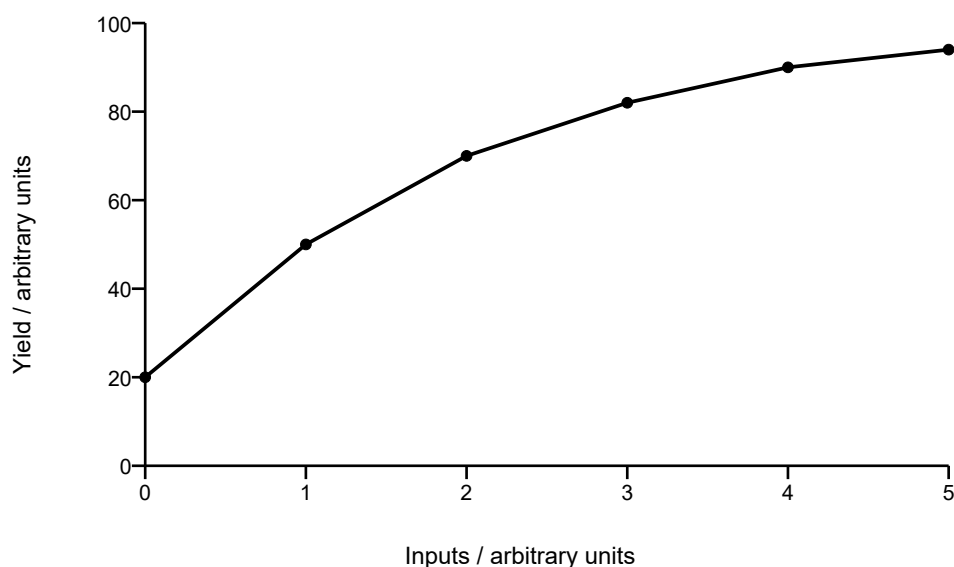
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07

Figure 1 shows the effect of increasing inputs (fertiliser, water and pesticides) on the yield of a crop.

Figure 1



07 . 1

Describe the relationship shown in Figure 1.

[2 marks]

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07 . 2

Calculate the extra yield produced by the second unit of input and by the fifth unit of input.

[2 marks]

Answer

units

07 . 3

Explain why, if enough land is available, spreading inputs evenly over a large area gives a greater total yield than concentrating them on a small area.

[2 marks]

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07 . 4

Suggest why inputs are nevertheless often used unevenly across the world.

[2 marks]

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07 . 5

Explain why intensive agriculture may be very productive but not very efficient.

[2 marks]

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08 . 1

Explain the difference between hard and soft release and state the types of species for which each is suitable.

[3 marks]

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08 . 2

Describe **five** features of a suitable release site.

[5 marks]

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08 . 3

Explain the importance of post-release monitoring.

[2 marks]

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09 . 1

Explain why the ecosystem services provided by other species are often given a low conservation priority.

[2 marks]

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09 . 2

Explain how soil organisms provide services that humans depend on.

[3 marks]

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09 . 3

Explain why pollination by insects is more useful than wind pollination to plants growing in forests.

[2 marks]

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09 . 4

Explain how seed dispersal by animals increases the survival of a plant's offspring.

[3 marks]

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10 . 1

Evaluate the extent to which the principles of the circular economy can be applied to **three** of the following: water supplies, mineral supplies, waste management, pollution control, energy supplies, food production.

[9 marks]

Extra space

10 . 2 State what is meant by **industrial symbiosis**.

[1 mark]

10

**Write an essay on one of the following topics.**

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.

**[25 marks]**

**OR**

$$\boxed{1} \boxed{1} . \boxed{2}$$

Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.

**[25 marks]**

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.



Do not write outside the box

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Do not write outside the box

**END OF QUESTIONS**